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BCS 201: Introduction to Pathology - Disease, Injury and Repair

The core concepts of disease processes for students entering the clinical sciences

FACULTY PRIMER

Faculty Primer

Faculty of Basic Clinical Sciences
Chemical Pathology | 200L

Written for students, by students.

Practical notes, worked examples, practice questions and a revision plan.

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Independent study support | Not an official UNIZIK publication

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Why this pack exists

Pathology explains what goes wrong and why. This pack introduces cell injury, inflammation, repair, haemodynamics and neoplasia as a connected story, with the vocabulary you need for later clinical courses and a method for describing any pathological process.

This is a student-written study companion, not a reproduced lecture handout or a leaked examination paper. Use it to organise the topics you are taught, practise the kind of thinking your course rewards, and walk into class with better questions.

What you will be able to do

- Distinguish reversible from irreversible cell injury and name common causes.
- Describe the stages and cells of acute inflammation and its clinical signs.
- Explain tissue repair, regeneration and fibrosis.
- Define neoplasia and distinguish benign from malignant tumours.

How to use this pack

1. Skim the whole pack once in 20 minutes so you know what is inside.
2. Work through one section at a time with your lecture note and recommended text beside you.
3. Attempt the practice questions before reading the answers. Mark what you missed.
4. Use the final checklist to plan your revision week, one tick per study session.

> **Free to download, always**
> This pack is free. There is no fee, no subscription and no payment step before download, and the full PDF is not watermarked or locked. Share the link with a coursemate instead of forwarding the file.

> **Confirm your official course outline**
> Course codes, credit units and outlines can differ by department and session. Treat this pack as supplementary support and confirm the current outline with your lecturer, course rep or departmental handbook before an assessment.

Cell injury and adaptation

Cells respond to stress by adapting; when adaptation fails, injury follows.

- Adaptations: hypertrophy, hyperplasia, atrophy and metaplasia.
- Reversible injury: cellular swelling and fatty change.
- Irreversible injury: mitochondrial failure, membrane damage and cell death.
- Necrosis is uncontrolled cell death with inflammation; apoptosis is programmed and clean.
- Common causes: hypoxia, toxins, infection, physical trauma, immune reactions and genetic defects.
- Free radicals and calcium overload are recurring mechanisms of damage.

> **Link mechanism to appearance**
> Exam answers improve immediately when you connect a mechanism - such as ATP depletion - to its morphological consequence, like swelling.

Inflammation and repair

Inflammation is a protective response that can also cause harm when excessive or prolonged.

- Cardinal signs: redness, heat, swelling, pain and loss of function.
- Acute inflammation: vasodilation, increased permeability and neutrophil recruitment.
- Chemical mediators: histamine, prostaglandins, leukotrienes and cytokines.
- Chronic inflammation: mononuclear cells, tissue destruction and attempted repair occurring together.
- Granulation tissue forms the scaffold for healing.
- Repair may be by regeneration of the same tissue or by fibrosis with scar formation.

Haemodynamics and neoplasia

Circulatory disturbances and uncontrolled growth account for a large share of clinical disease.

- Oedema results from increased hydrostatic pressure, reduced oncotic pressure, lymphatic obstruction or sodium retention.
- Thrombosis is clot formation in an intact vessel; embolism is material travelling in the bloodstream.
- Infarction is tissue death from ischaemia.
- Neoplasia is new, autonomous growth; the cell escapes normal growth control.
- Benign tumours are well circumscribed and do not metastasise; malignant tumours invade and metastasise.
- Grading describes differentiation; staging describes spread.

What pathology studies

Pathology studies the causes, mechanisms and effects of disease, answering four questions: cause (aetiology), mechanism (pathogenesis), structural change (morphology) and functional consequence (clinical significance).

Disease is the failure of homeostasis. Cells respond to injury by adapting, degenerating reversibly, or dying. That repertoire is limited - hypertrophy, atrophy, hyperplasia, metaplasia, necrosis, apoptosis - which is why a small set of concepts explains a very large number of conditions.

Reversible injury produces swelling and fatty change with function intact once the cause is removed; irreversible injury produces cell death, which the body must repair or replace with scar.

Inflammation, repair and neoplasia